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Mayiladuturai District 2026

COMMON QUARTERLY EXAMINATION - 2026

Standard XII

PHYSICS

Reg No. | | | | | | |

Time : 3.00 hrs *Marks : 70*

Part - I

I. Choose the correct answer: 15 x 1 = 15

1. The frequency order for gamma rays (B), X-rays (A), UV rays (C) is
   a) B > A > C b) A > B > C c) C > B > A d) A > C > B
2. If voltage applied on a capacitor is increased from V to 2V, choose the correct conclusion
   a) Both Q and C remain same b) Q remains the same, C is doubled
   c) C remains same, Q doubled d) Q is doubled, C doubled
3. A car battery of emf 12V and internal resistance $5 \times 10^{-2}\ \Omega$, receive a current of 60 A, from external source, the terminal potential difference of battery is
   a) 12 V b) 9 V c) 15 V d) 20 V
4. A toaster operating at 240 V has a resistance of 120 Ω. The power is
   a) 400 W b) 2 W c) 480 W d) 240 W
5. In a series RL circuit, the resistance and inductive reactance are the same. Then the phase difference between the voltage and current is
   a) $\frac{\pi}{4}$ b) $\frac{\pi}{2}$ c) $\frac{\pi}{6}$ d) zero
6. For light incident from air on a slab of refractive index 2, the maximum possible angle of refraction is
   a) 30° b) 45° c) 60° d) 90°
7. A non-conducting charged ring of charge 'q', mass 'm' and radius 'r' is rotated with constant angular speed ω. Find the ratio of its magnetic moment with angular momentum is
   a) $\frac{q}{m}$ b) $\frac{2q}{m}$ c) $\frac{q}{2m}$ d) $\frac{q}{4m}$
8. The number of electrons in one coulomb of negative charge is
   a) 6.25×10^{10} electrons b) 6.25×10^{-10} electrons
   c) 6.25×10^{-18} electrons d) 6.25×10^{18} electrons
9. The speed of light in an isotropic medium depends on
   a) it's intensity b) it's wavelength
   c) the nature of propagation d) the motion of the source w.r.t. medium

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force on it is

a) $q(\vec{V}\times\vec{B})$ b) $q\vec{E}+q(\vec{V}\times\vec{B})$

c) $q\vec{E}+q(\vec{B}\times\vec{V})$ d) $q\vec{B}+q(\vec{E}\times\vec{V})$

11. The flux linked with a coil at any instant 't' is given by $\phi_B = 10t^2 - 50t + 250$. The induced emf at t = 3 s is

a) −190 V b) −10 V c) 10 V d) 190 V

12. A radiation of energy E falls normally on a perfectly reflecting surface. The momentum transferred to the surface is

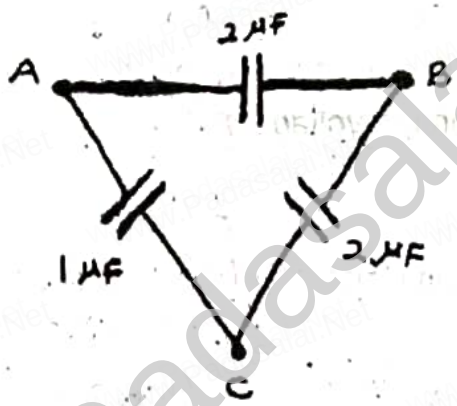
$P_i = \frac{E}{c}$

$P_r = -\frac{E}{c}$

$P_i - P_f = 2E/c$

a) $\frac{E}{C}$ b) $\frac{2E}{C}$ c) EC d) $\frac{E}{C^2}$

13. Three capacitors are connected in triangle as shown in the figure. The equivalent capacitance between the point A and C is



a) 1 μF b) 2 μF c) 3 μF d) $\frac{1}{4}$ μF

14. The vertical component of Earth's magnetic field at a place is equal to the horizontal component. What is the value of angle of dip at this place?

a) 30° b) 45° c) 60° d) 90°

15. The temperature co-efficient of resistance of a wire is 0.00125 per °C. At 300 K, its resistance is 1 Ω. The resistance of the wire will be 2 Ω at

a) 1154 K b) 1100 K c) 1400 K d) 1127 K

$R_T = R_0(1+\alpha T)$

$T = 300 - 273 = 27°C$

$R_1 = R_0(1 + 0.00125 \times 27)$

$= 1.03375$

20 × 1 = 20°C

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Part - II

II. Answer any Six questions. (Q.No.24 is compulsory) 6 x 2 = 12

16. Define electrostatic potential.
17. What is displacement current?
18. State Fleming's left hand rule.
19. How will you define Q-Factor?
20. Define current density.
21. What is meant by hysteresis?
22. State Ampere's circuital law.
23. Define electric resonance.
24. Compute the speed of the electromagnetic wave in a medium, if the amplitude of electric and magnetic fields are 3×10^4 NC^{-1} and 2×10^{-4} T respectively.

Part - III

III. Answer any Six questions. (Q.No.33 is compulsory) 6 x 3 = 18

25. State and explain Kirchhoff's rules.
26. Distinguish between Coulomb force and Gravitational force.
27. State the applications of Seeback effect.
28. Give the properties of dia, para and ferro magnetism.
29. Find out the phase relationship between voltage and current in a pure resistor AC circuit.
30. Derive the relation between F and R for a spherical mirror.
31. Write short notes on
    a) X-ray
    b) Microwave
32. A parallel plate capacitor has square plate of side 5 cm and separated by a distance of 1 mm. Calculate the capacitance of this capacitor.
33. Two materials X and Y are magnitised whose intensity of magnetisation are 500 Am^{-1} and 2000 Am^{-1} respectively. If the magnetising field is 1000 Am^{-1}, then which one among these materials can be easily magnetized?

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Part - IV

IV. Answer all the questions. 5 x 5 = 25

34. a) Derive the expression for resultant capacitance, when the capacitors are connected in series and in parallel.

(OR)

b) Derive the mirror equation.

35. a) Describe the microscopic model of current and obtain general form of Ohm's law.

(OR)

b) i) State and explain Biot-Savart's law.

ii) Compute the magnitude of the magnetic field of a long, straight wire carrying on a current of 1 A at a distance of 1 m from it.

36. a) i) How is a galvanometer converted into an ammeter?

ii) Is an ammeter connected in series or parallel in a circuit? Why?

(OR)

b) What is emission spectrum? Classify with example.

37. a) Explain the construction and working of transformer.

(OR)

b) Explain in detail the construction and working of a Van de Graff generator.

38. a) Show that the mutual inductance between a pair of coils is same. ($M_{12} = M_{21}$)

(OR)

b) How the emf of two cells are compared using potentiometer?

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